LAGRANGE COLLISIONS AND COVER RELATIONS FOR RATIONAL DYCK PATHS

ALEX CHENGYU LI

Formalization status. An accompanying Lean 4 development kernel-checks every numbered result and displayed formula identified in the manuscript map, together with all exact certificates described in Section 8. The publication root is premise-free and derives the snake-graph matching formula and the periodic Lagrange formula from Mathlib foundations.

Abstract. Schiffler associated matching and Lagrange orders to rational Dyck paths and posed two problems about their equality and cover relations. We first classify the band-graph isomorphism classes at a fixed coprime endpoint: every class is an orbit of an explicit reversal involution and therefore has size at most two. We then give two paths in $\mathcal{D}(17,9)$ with the same exact Lagrange value but in different band-graph orbits, answering the equality question negatively. Two independent exact enumerations through total length 25 show that the length-26 example is computationally minimal.

For the cover problem, we attach exact matrix intervals to rational-Dyck prefixes. A best-first traversal emits complete equality levels in increasing order for both scores; equivalently, a pair is a cover precisely when a finite prefix antichain certifies all path cylinders outside its open score interval. We supplement this global characterization with structural matching results: the parity of every matching score is determined by the endpoint, every score gap of two is a cover, a local interchange has an exact four-case continuant formula, and the first n matching levels in $\mathcal{D}(n,n-1)$ are explicit singletons. The last result yields covers at Hamming distance $2n-6$, ruling out any uniformly bounded local-move catalogue. All finite certificates use integer arithmetic or reduced rational squares.

1. The problems and main results

Let $a>b>0$ be coprime. The set $\mathcal{D}(a,b)$ consists of the words with a letters R and b letters U for which every prefix v satisfies

$$a\,\#_{\mathrm{U}}(v)\le b\,\#_{\mathrm{R}}(v).$$

Date: August 2026.

2020 *Mathematics Subject Classification.* Primary 05A15, 06A07; Secondary 11A55, 05C70.

Key words and phrases. lattice paths, continuants, Lagrange spectrum, band graphs, cover relations, exact certificates.

Schiffler [5] associates two strict partial orders to these paths. The matching score is the numerator of a finite continued fraction, and the Lagrange score is the Lagrange number of an associated periodic continued fraction. The underlying snake and band graphs belong to the graph calculus developed by Çanakçı and Schiffler [2]. Problem 6.3 of [5] asks whether equality of Lagrange values for two paths in the same $\mathcal{D}(a,b)$ implies isomorphism of their band graphs. Problem 6.5 asks for a path description of the cover relations in either order. Apruzzese and Cong [1] prove the common unique maximum and the supremum of all Lagrange values; they reiterate that comparison criteria alone do not exclude an intermediate score level.

The first structural point is that band-graph isomorphism itself has a simple path description. Reverse the cyclic step word of ω and rotate it to the unique member of $\mathcal{D}(a,b)$ in that rotation class; call the result $\tau(\omega)$. The existence and uniqueness are the coprime cycle lemma, recalled in Section 3.

Theorem 1.1 (Band-graph classes)**.** *For $\omega,\omega' \in \mathcal{D}(a,b)$,*

$$\overline{G}(\omega) \cong \overline{G}(\omega') \quad \iff \quad \omega' = \omega \text{ or } \omega' = \tau(\omega).$$

In particular, every band-graph isomorphism class has size at most two.

This makes the equality question concrete: one must decide whether a Lagrange level can meet two different τ-orbits. Our first counterexample does so.

Theorem 1.2 (Counterexample to Problem 6.3)**.** *There are paths $\omega,\omega' \in \mathcal{D}(17,9)$ such that*

$$L(\omega) = L(\omega') \quad \textit{but} \quad \overline{G}(\omega) \not\cong \overline{G}(\omega').$$

They can be chosen with step words

$$\omega = \texttt{RRRRRRURRURRUURURRURRRUURU},$$

$$\omega' = \texttt{RRRRRRURUURURRURRRURRURRUU}.$$

The equality and nonisomorphism are finite exact statements and do not depend on floating-point approximations.

Our answer to the cover problem is constructive and global. Explicit lower and upper score bounds on every path-prefix cylinder yield both a score-ordered generator and a locally checkable prefix-antichain certificate for any proposed cover. No bounded catalogue of local swaps is required.

Theorem 1.3 (Constructive cover characterization)**.** *For every coprime $a > b > 0$, there is an exact best-first traversal of the rational-Dyck prefix tree that emits the complete matching-score levels, and also the complete Lagrange-score levels, in strictly increasing batches. For either order, the cover relation is the union of the complete bipartite relations between consecutive emitted batches.*

Equivalently, paths $x < y$ form a cover if and only if there is a finite prefix antichain partitioning $\mathcal{D}(a,b)$ such that the score interval of every antichain *cylinder is certified to lie at or below the score of x, or at or above the score of y.*

The bounds and the certificate criterion are given explicitly in Section 6. The matching order also admits useful uniform structure.

Theorem 1.4 (Endpoint parity). *For every $\omega \in \mathcal{D}(a,b)$,*

$$M(\omega) \equiv 0 \pmod 2 \iff 3 \mid (a-b).$$

Consequently all matching scores at a fixed endpoint have the same parity, and any two realized scores differing by two are consecutive levels.

Finally, the next theorem gives a closed initial segment and shows why the global form of Theorem 1.3 is necessary. For a path ω, write $A_E(\omega)$ for its word in the adjacency blocks D, E of Section 2.

Theorem 1.5 (Initial matching levels and unbounded nonlocality). *For every $n \geq 4$, the paths*

$$X_n = \mathrm{RRR(UR)}^{n-3}\mathrm{UU}, \qquad Y_n = \mathrm{RR(UR)}^{n-4}\mathrm{RUURU}$$

form a matching cover in $\mathcal{D}(n, n-1)$ and have Hamming distance $2n-6$. More precisely, the first n distinct matching levels are singletons. The minimum is the path with block word ED^{2n-3}; the next $n-2$ paths have block words

$$ED^{2r}ED^{2n-5-2r}E, \qquad 0 \leq r \leq n-3,$$

listed in decreasing order of r; and Y_n is the unique path at the next level. In this list X_n is the level immediately below Y_n.

Theorem 1.1 and the parity and initial-level theorems are uniform structural results. Their proofs use, respectively, the coprime cycle lemma, a two-by-two calculation over $\mathbb{F}_2$, and an elementary Pell-continuant extremal lemma.

2. Background and exact matrix model

For a path ω, form a finite coefficient word $A(\omega)$ from adjacent steps: an equal pair contributes the two coefficients $1, 1$, while an unequal pair contributes the coefficient 2. Schiffler's matching score $M(\omega)$ is the numerator of the continued fraction with coefficient word $A(\omega)$. Closing the step word cyclically adds one unequal edge, and the corresponding periodic continued fraction has Lagrange number $L(\omega)$. The two strict orders considered here are the pullbacks of the usual order on these scores:

$$\omega <_M \eta \iff M(\omega) < M(\eta), \qquad \omega <_L \eta \iff L(\omega) < L(\eta).$$

Paths with equal scores are therefore incomparable. A cover is a pair of paths in two consecutive distinct realized score levels.

Put

$$T(c) = \begin{pmatrix} c & 1 \\ 1 & 0 \end{pmatrix}, \qquad D = T(2) = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad E = T(1)^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

For adjacent steps of a path, assign the block E when the steps agree and D when they differ. If the product of the $a+b-1$ adjacency blocks is $P(\omega)$, then

$$M(\omega) = P(\omega)_{11}.$$

This is exactly Schiffler's rule replacing an equal pair by the digits $1, 1$ and an unequal pair by the digit 2.

For completeness, this identity can also be obtained directly from the graph. Expose the terminal edge of a square snake and keep two states: p is the number of perfect matchings, and q is the number of matchings that leave exactly the two terminal vertices unmatched. Attaching one square gives

$$\begin{aligned} (p, q) &\longmapsto (p+q, p) && \text{for a straight continuation,} \\ (p, q) &\longmapsto (p+q, q) && \text{at a turn.} \end{aligned}$$

Indeed, the two new vertices are matched either to each other or to the two vertices of the shared edge. Starting from $(2, 1)$ for one square, the two half-tile attachments associated with an adjacent pair of path steps act by E when the steps agree and by D when they differ. Thus the first state is $P(\omega)_{11}$ and counts the actual edge subsets that meet every graph vertex exactly once.

Close the step word cyclically. Every path in $\mathcal{D}(a, b)$ begins with R and ends with U, so the closing edge is unequal. Thus the cyclic matrix is

$$F(\omega) = DP(\omega).$$

The number of unequal edges in the closed binary word is even. Hence $\det F(\omega) = 1$. For every cyclic digit cut, write the corresponding product as

$$F_i = \begin{pmatrix} p_i & r_i \\ q_i & s_i \end{pmatrix}.$$

All these products have the same trace. The two fixed points of the associated linear fractional transformation differ by

$$\frac{\sqrt{(p_i - s_i)^2 + 4r_i q_i}}{q_i} = \frac{\sqrt{\operatorname{tr}(F)^2 - 4}}{q_i}.$$

To relate this calculation to the standard Lagrange number, let α_i be the positive purely periodic continued fraction at the ith cut. Reversing the period transposes its matrix product. If α_i' is the negative conjugate root, then $-1/\alpha_i'$ is the positive fixed point of the reversed period. Hence the backward tail is $-\alpha_i'$, and the usual forward-plus- backward approximation coefficient is precisely $\alpha_i - \alpha_i'$. Taking the maximum over the finitely many cyclic cuts therefore gives

$$L(\omega)^2 = \frac{\operatorname{tr}(F(\omega))^2 - 4}{q_{\min}(\omega)^2}, \qquad q_{\min}(\omega) = \min_i q_i. \tag{1}$$

Equation (1) gives an exact reduced rational comparison for the Lagrange order, since all Lagrange values here are positive.

3. Band-graph classes

Close the step word of ω cyclically. Encode an equal cyclic edge by the pair $1, 1$ and an unequal edge by 2. Since every path begins with R and ends with U, the closing edge contributes a leading 2, and the resulting cyclic coefficient word is exactly

$$C(\omega) = (2, A(\omega)).$$

We regard $C(\omega)$ with its intrinsic parsing into the blocks $(1, 1)$ and (2). Equivalently, this parsing is recovered from the flattened digit word: every cyclic run of 1's has even length, and every 2 is a singleton block.

Lemma 3.1 (Recovery from the cyclic code)**.** *The cyclic equality/difference word of the steps, equivalently the cyclic coefficient word $C(\omega)$, determines the cyclic step word of ω up to global complementation. If two paths in the same $\mathcal{D}(a, b)$ have either of these codes related by a rotation or reversal, then their cyclic step words are related by a rotation or reversal, without complementation.*

Proof. Pair the 1's in each cyclic run of $C(\omega)$ and replace every pair by an equal edge; replace each 2 by an unequal edge. This recovers the cyclic equality/difference word of the steps. The parsing commutes with cyclic rotation and reversal, because the blocks $(1, 1)$ and (2) are palindromic. After one step is chosen, all remaining steps are forced. The other initial choice complements every step. Global complementation exchanges the numbers of R's and U's, and hence cannot take one word with a copies of R and b copies of U to another because $a \neq b$. □

We next record the coprime cycle lemma in the form needed here. Give an R-step weight b and a U-step weight $-a$.

Lemma 3.2 (Unique admissible rotation)**.** *Every cyclic word with a copies of* R *and b copies of* U *has a unique rotation in $\mathcal{D}(a, b)$.*

Proof. The total weight is zero. No two proper cyclic prefix sums are equal: the intervening proper segment would have $b\,\#_{\mathrm{R}} = a\,\#_{\mathrm{U}}$, and coprimality would force it to contain all a right steps and all b up steps. Thus the cyclic prefix sums are distinct. Rotating immediately after their unique minimum makes every proper prefix sum positive, while any other rotation has a negative prefix ending at that minimum. This is precisely the defining prefix inequality. □

Let $\tau(\omega)$ be the unique admissible rotation of the reversed step word. Lemma 3.2 shows at once that τ is an involution.

Lemma 3.3 (Intrinsic tile cycle)**.** *The abstract graph $\overline{G}(\omega)$ determines the cyclic equality/difference word of the steps up to rotation and reversal. Conversely, a rotation or reversal of that word induces an isomorphism of the band graphs.*

Proof. Schiffler's construction places one square at every step midpoint and every interior path vertex, adds the closing square, and glues the two terminal edges; see [2, 5]. Before the final gluing, the square centers form a monotone width-one ribbon. It contains no 2×2 block, so every chordless four-cycle in its one-skeleton is the boundary of a single square. After gluing, a new four-cycle using the glued edge would give a three-edge path between its endpoints in the open ribbon. The only such paths are the other three sides of the two squares incident with that edge; the alternative route traverses the remaining ribbon, which has at least six tiles. Thus the chordless four-cycles of $\overline{G}(\omega)$ are precisely its $2(a+b)$ tiles, and their shared-edge intersection graph is a cycle.

The tile cycle is bipartite. One bipartition class consists of the step midpoint squares; on each of them the two shared edges are opposite. The other class consists of the path-vertex squares. On such a square the shared edges are opposite when the two cyclically adjacent steps agree and adjacent when they differ. Since the closed step word contains both letters, the latter class contains an adjacent-edge square and is therefore distinguished from the midpoint class by the abstract graph. Reading it around the tile cycle recovers the cyclic equality/difference word up to the dihedral symmetries of that cycle. Reversing this construction proves the converse. □

Proof of Theorem 1.1. Lemma 3.3 says that the band graphs are isomorphic if and only if the cyclic equality/difference codes are related dihedrally. By Lemma 3.1, their step words are then related by rotation or reversal. Lemma 3.2 leaves only ω itself in the rotation class and $\tau(\omega)$ in the reversed class. The converse follows directly from the construction. □

The classification also gives a closed count. Put $N = a + b$ and define

$$R_{a,b} = \begin{cases} \dbinom{(N-1)/2}{\lfloor a/2 \rfloor}, & N \text{ odd}, \\ \dbinom{(N-2)/2}{(a-1)/2}, & N \text{ even}. \end{cases}$$

In the second case a and b are both odd. The usual coprime cycle lemma gives $|\mathcal{D}(a,b)| = \binom{N}{a}/N$. Aperiodicity follows from coprimality. When N is odd, a reflection fixes one position and pairs the remaining positions; when N is even, only the reflections fixing two positions can preserve the odd number a of right steps. Counting the fixed binary necklaces gives $R_{a,b}$. Burnside's lemma for the involution τ now yields the following.

Corollary 3.4 (Number of band-graph classes)**.** *The number of band-graph isomorphism classes in $\mathcal{D}(a,b)$ is*

$$\frac{1}{2}\left(\frac{1}{N}\binom{N}{a}+R_{a,b}\right).$$

For $\mathcal{D}(n,n-1)$ this becomes

$$\frac{1}{2}\left(\mathrm{Cat}_{n-1}+\binom{n-1}{\lfloor (n-1)/2\rfloor}\right).$$

4. The counterexample

Direct prefix checking places the two words of Theorem 1.2 in $\mathcal{D}(17,9)$. Their adjacency digit words both have 37 digits and their cyclic words both have 38 digits. Exact matrix multiplication gives, for each path,

$$\begin{aligned} \operatorname{tr}(F) &= 20034079647, \\ \operatorname{tr}(F)^2-4 &= 401364347302339644605, \\ q_{\min} &= 6262579207. \end{aligned}$$

Therefore

$$L(\omega)^2 = L(\omega')^2 = \frac{401364347302339644605}{39219898323948748849}. \tag{2}$$

It remains to separate the unlabelled band graphs. We use an invariant that is recovered from the abstract graph. In each of the two band graphs, the chordless 4-cycles are exactly the 52 square tiles. Their edge-intersection graph is a 52-cycle. At each tile, record 1 if the two edges shared with its two neighboring tiles are adjacent, and 0 if they are opposite. An abstract graph isomorphism must rotate or reverse this cyclic word. The lexicographically least dihedral representatives for the two graphs are

$$\texttt{0000000000010100010100010001010100010100000100010101},$$
$$\texttt{0000000000010001000101000101000001010001010100010101}.$$

They differ, so the band graphs are not isomorphic. This proves Theorem 1.2.

For reference, the corresponding least dihedral representatives of the cyclic coefficient words are

$$\texttt{11111111112211221121122211221111211222},$$
$$\texttt{11111111112112112211221111221122211222}.$$

Proposition 4.1 (Computational minimality through length 25)**.** *No coprime endpoint of total length at most* 25 *contains a counterexample to Problem 6.3. Thus the example above is computationally minimal in total length.*

Exact finite verification. For every coprime pair $a>b>0$ with $a+b\le 25$, rational-Dyck words are generated recursively, retaining precisely the one-step extensions satisfying the prefix inequality. This gives all 99 endpoints and all

934,635 paths in the stated range. For each path, equation (1) is evaluated with arbitrary-precision integer matrices. Equality of two Lagrange squares is decided by exact cross multiplication, and the band class is represented by the least rotation among the cyclic turn word and its reversal. Grouping by the exact Lagrange square produces 471,455 score classes; every class is contained in one band-graph class.

For each endpoint, the number generated is also checked against the coprime cycle-lemma count $\binom{a+b}{a}/(a+b)$. Hence the recursion omits no path in the stated range.

An independent C++ implementation reproduces every endpoint, path, and score-class count through length 25 and again verifies that no score class meets two band classes. Its unsigned 128-bit arithmetic is exact on this range: the sum of the cyclic coefficients is at most 50, so every continuant is at most Fib_{51} and every squared quantity is far below 2^{128}. A third exact implementation verifies the displayed length-26 pair and constructs its two abstract band graphs directly. The agreement of the two exhaustive enumerations proves the proposition. □

5. Structural matching covers

We first prove that the parity of the matching score does not depend on the path. Work over $\mathbb{F}_2$ and put $F = T(1)$ and $S = T(2)$. Direct multiplication gives

$$F^3 = I, \qquad E = F^2 = F^{-1}, \qquad SFS = F^{-1}.$$

Write a path by its nonempty runs as

$$\mathrm{R}^{r_1}\mathrm{U}^{u_1}\cdots\mathrm{R}^{r_k}\mathrm{U}^{u_k}.$$

Its adjacency product modulo two is

$$E^{r_1-1}SE^{u_1-1}S\cdots SE^{r_k-1}SE^{u_k-1}.$$

Moving every occurrence of S to the right by $SF^j = F^{-j}S$, and using $\sum r_i = a$ and $\sum u_i = b$, reduces this product to

$$P(\omega) \equiv F^{b-a}S \pmod 2. \tag{3}$$

The upper-left entry of F^jS is zero precisely when $j \equiv 0 \pmod 3$. This proves Theorem 1.4.

Corollary 5.1 (Unit-gap covers)**.** *If $x, y \in \mathcal{D}(a,b)$ and $M(y) = M(x)+2$, then y covers x in the matching order.*

Proof. All realized matching scores in this endpoint class have the same parity by Theorem 1.4. No integer of that parity lies strictly between $M(x)$ and $M(y)$. □

The local interchange proposed after Problem 6.5 also has a closed score formula, although it does not have a uniform direction. Suppose two valid paths differ by an interior interchange

$$\mathrm{old} = \alpha\mathrm{RU}\beta, \qquad \mathrm{new} = \alpha\mathrm{UR}\beta.$$

Let ℓ and r be the steps immediately before and after the interchanged pair. Write the matrix products strictly before and after the three affected adjacency blocks as

$$P = \begin{pmatrix} p & r_0 \\ q & s \end{pmatrix}, \qquad Q = \begin{pmatrix} u & v \\ w & z \end{pmatrix}.$$

Proposition 5.2 (Exact local-interchange formula)**.** *The matching-score difference is given by the following four cases.*

(ℓ, r)	*old blocks*	*new blocks*	$M(\text{old}) - M(\text{new})$
(R, R)	EDD	DDE	$2(r_0u - pw)$
(R, U)	EDE	DDD	$2(pw + r_0u + r_0w) > 0$
(U, R)	DDD	EDE	$-2(pw + r_0u + r_0w) < 0$
(U, U)	DDE	EDD	$2(pw - r_0u)$

Proof. Only the first row of P and the first column of Q contribute to the upper-left entry of the complete product. Substitution of

$$EDE - DDD = \begin{pmatrix} 0 & 2 \\ 2 & 2 \end{pmatrix}, \qquad EDD - DDE = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$$

gives the four displayed expressions. All entries of P and Q that occur in the middle two cases are nonnegative, with $p, u > 0$. The sum $pw + r_0u + r_0w$ could vanish only if both exterior products were empty; the entire path would then contain two right and two up steps, contrary to $a > b$. Hence the two displayed signs are strict. ☐

Thus the two mixed-neighbor cases force opposite directions, while the equal-side cases compare the prefix and suffix continuant ratios. Combined with Corollary 5.1, the identities $|r_0u - pw| = 1$ or $pw + r_0u + r_0w = 1$ certify a matching cover without enumerating the remaining paths. The formula alone is not a complete cover criterion: larger differences can still be consecutive among the realized scores.

6. Path-prefix bounds and all cover relations

Fix a feasible prefix α of length $k \geq 1$. Let P_α be the product of its $k-1$ known adjacency blocks, and put $r = a + b - k$. Every unknown block B satisfies $D \leq B \leq E$ entrywise. Since all entries are nonnegative, multiplication preserves these inequalities.

6.1. **Matching bounds.** Every completion ω of α satisfies

$$\ell_M(\alpha) := (P_\alpha D^r)_{11} \leq M(\omega) \leq (P_\alpha E^r)_{11} =: u_M(\alpha). \tag{4}$$

6.2. **Lagrange bounds.** Trace monotonicity gives

$$T_-(\alpha) := \mathrm{tr}(DP_\alpha D^r) \le \mathrm{tr}(F(\omega)) \le \mathrm{tr}(DP_\alpha E^r) =: T_+(\alpha).$$

At the unshifted cut,

$$q_{\min}(\omega) \le q_0(\omega) \le (DP_\alpha E^r)_{21} =: Q_+(\alpha).$$

This yields the lower bound

$$\ell_L(\alpha) = \frac{\max\{0, T_-(\alpha)^2 - 4\}}{Q_+(\alpha)^2} \le L(\omega)^2. \tag{5}$$

Let $e(\alpha)$ be the number of equal adjacencies already present in the prefix. The completed cyclic digit word has at least $a + b + e(\alpha)$ digits. At any cut, its lower-left entry is a positive-digit continuant of all digits except the first. It is therefore at least the all-ones value $\mathrm{Fib}_{a+b+e(\alpha)}$, where $\mathrm{Fib}_0 = 0$, $\mathrm{Fib}_1 = 1$. Hence

$$L(\omega)^2 \le u_L(\alpha) := \frac{\max\{0, T_+(\alpha)^2 - 4\}}{\mathrm{Fib}^2_{a+b+e(\alpha)}}. \tag{6}$$

At a leaf, all four bounds are replaced by the corresponding exact score.

6.3. **Best-first level theorem.** The feasible prefixes form a finite rooted tree. Maintain a frontier antichain partitioning the not-yet-emitted leaves. For the matching traversal, key a nonleaf node by ℓ_M; for the Lagrange traversal, key it by ℓ_L. Key a leaf by its exact matching score or exact reduced Lagrange square. Repeatedly expand a minimum-key nonleaf node or emit a minimum-key leaf, with nonleaf nodes taking precedence at an equal key.

Lemma 6.1. *For either choice of score, leaves are emitted in nondecreasing exact score order, and every equality level is emitted as one complete batch.*

Proof. Suppose a leaf of score s is about to be emitted while an unemitted leaf of score $t < s$ remains. Exactly one current frontier prefix contains that leaf. Its key is a lower bound for all its completions, so its key is at most t and it has priority over the leaf of key s, a contradiction. The same argument with $t = s$, together with nonleaf-first tie handling, proves that the equality batch is complete before a larger leaf is emitted. □

Lemma 6.1 proves the first statement of Theorem 1.3. Indeed, in a strict order pulled back from a scalar score, a path in level A is covered by a path in level B exactly when A and B are consecutive distinct scalar levels.

6.4. **A direct cover certificate.** For a prefix α, let $I(\alpha) = [\ell(\alpha), u(\alpha)]$ be the applicable interval from (4), or from (5) and (6).

Proposition 6.2 (Prefix-antichain criterion)**.** *Let $x, y \in \mathcal{D}(a, b)$ have exact scores $s < t$ in either order. Then y covers x if and only if there is a finite* *prefix antichain $\mathcal{A}$ whose cylinders partition $\mathcal{D}(a,b)$ and such that every $\alpha \in \mathcal{A}$ satisfies*

$$u(\alpha) \le s \qquad \textit{or} \qquad \ell(\alpha) \ge t.$$

Proof. If the antichain exists, the score of every path lies outside (s,t), so no third level occurs between x and y. Conversely, refine every prefix whose interval meets (s,t). The tree is finite. At a leaf the interval is the singleton exact score. If $x < y$ is a cover, every leaf is at most s or at least t, so the refinement terminates with the required antichain. □

This proves the second statement of Theorem 1.3. Notice that Proposition 6.2 is a verifiable global interval-exclusion certificate. It does not merely recompute the two endpoint scores.

There is an equivalent deterministic formulation. For exact scores $s < t$, define $\mathsf{C}_{s,t}(\alpha)$ recursively: it is true when $u(\alpha) \le s$ or $\ell(\alpha) \ge t$; it is false at an otherwise undecided leaf; and at an otherwise undecided nonleaf prefix it is the conjunction of $\mathsf{C}_{s,t}$ over all feasible one-step extensions. Then

$$x \lessdot y \iff \mathsf{C}_{s,t}(\mathrm{R}) \text{ is true.}$$

Thus the criterion does not assume a cover oracle. A successful recursion returns the terminal true prefixes as its antichain certificate; a failed recursion returns an intermediate path.

Remark 6.3 (Certificate size)**.** The full set of leaves is the worst-case certificate in the necessity proof. Thus Theorem 1.3 gives an exact global characterization but does not imply a uniformly short certificate or a finite catalogue of local swaps. For scale, the matching cover $X_{13} \lessdot Y_{13}$ has a certificate with 122 terminal prefixes after 143 expansions, while $\mathcal{D}(13,12)$ has 208,012 paths.

7. An unbounded-distance matching-cover family

We give the proof of Theorem 1.5. Let

$$P_{-1} = 1, \quad P_0 = 0, \quad P_1 = 1, \qquad P_{j+1} = 2P_j + P_{j-1}.$$

Then

$$D^j = \begin{pmatrix} P_{j+1} & P_j \\ P_j & P_{j-1} \end{pmatrix}.$$

Put $m = 2n-2$ and $N = m-1$. If an adjacency-block word has initial block E and exactly two further E blocks at positions $1 \le i < j \le m-1$, expansion of $E = D + \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ gives

$$M(i,j) = P_{m+1} + f(i) + f(j) + P_i P_{j-i-2} P_{m-1-j}, \qquad f(k) = P_k P_{N-k}. \tag{7}$$

The convention $P_{-1} = 1$ covers adjacent positions.

Extremal continuant rearrangements such as Ramharter's [4] compare permutations of a fixed digit multiset. Here the paired-1 constraint, fixed endpoints, and rational-Dyck prefix condition change the admissible family, so we use the following self-contained Pell comparison.

The required Pell comparison is the following elementary lemma.

Lemma 7.1. *For odd $N \geq 5$ and $1 \leq k \leq N-1$,*

$$f(k) \leq f(1), \qquad f(k) \geq f(2),$$

with equality in the second inequality only at $k = 2, N-2$. Away from those two positions, $f(k) \geq f(4)$.

Proof. Write $N = 2s+1$ and use $f(k) = f(N-k)$. For $1 \leq k \leq s-1$, the determinant identities

$$f(k+2) - f(k) = 2(-1)^k P_{N-2k-2}, \qquad f(k) - f(k+1) = (-1)^{k+1} P_{N-2k-1}.$$

have positive Pell numbers on their right-hand sides. Hence the even subsequence increases from $f(2)$ and the odd subsequence decreases from $f(1)$ up to the midpoint. The second identity interlaces them: an even term is smaller than the next odd term, while an odd term is larger than the next even term. This proves the stated maximum and minimum, including the equality cases. Every even term other than $f(2)$ is at least $f(4)$, and every odd k with $3 \leq k \leq s-1$ satisfies $f(k) > f(k+1) \geq f(4)$. If s is odd, symmetry gives $f(s) = f(s+1) \geq f(4)$; and $f(1) \geq f(4)$ follows from the maximum assertion. Symmetry supplies the other half. $\square$

We first verify the path conditions explicitly. At the jth up-step in the alternating part of X_n, where $1 \leq j \leq n-3$, the numbers of right and up steps are $j+2$ and j; hence the prefix margin is

$$(n-1)(j+2) - nj = 2n - 2 - j > 0.$$

The two final up-steps have margins n and 0. For Y_n, the jth up-step in the alternating part has counts $j+1, j$ and margin $n-1-j > 0$ for $1 \leq j \leq n-4$; its three remaining up-steps have margins $n+1, 1, 0$. Thus both displayed words lie in $\mathcal{D}(n, n-1)$.

Moreover every path in this set begins with RR: the alternative prefix RU would require $n \leq n-1$. Every such path ends with U, because deleting a final R would leave all $n-1$ up-steps but only $n-1$ right-steps. There are $2n-2$ adjacency blocks, while a binary word beginning in R and ending in U has an odd number of unequal adjacencies. Its block word therefore begins with E and has an odd number of E blocks.

The block words of the two displayed paths are

$$A(X_n) = EED^{2n-5}E, \qquad A(Y_n) = ED^{2n-8}EDED^2.$$

Thus

$$M(X_n) = P_{m+1} + f(1), \qquad M(Y_n) = P_{m+1} + f(2) + f(4).$$

Their difference is positive because, with $h = m - 5 \geq 1$,

$$M(Y_n) - M(X_n) = 12P_h - P_{h+1} > 0.$$

Indeed $P_{h+1} = 2P_h + P_{h-1} < 3P_h$.

Now classify an arbitrary path by its odd number of equal adjacency blocks. A path with one E has score below $M(X_n)$. For a three-E word ending in E, equation (7) and Lemma 7.1 give score at most $M(X_n)$. For a three-E word ending in D, the same lemma gives score at least $M(Y_n)$, except apparently for positions $\{2, m-3\}$. In that case the interaction term in (7) gives

$$M(2, m-3) - M(Y_n) = 2P_{m-7} > 0.$$

Finally, in a word with at least five E blocks, keep the initial E and any two nonfinal E blocks and replace every other E by D. The result is a three-E word ending in D, and at least one nonfinal replacement is strict. Indeed, for nonempty products P, Q on the two sides of a replaced block,

$$\big(P(E-D)Q\big)_{11} = P_{12}Q_{21} > 0.$$

Hence the original score is above $M(Y_n)$. These cases exclude the whole open interval and prove the cover assertion. Direct comparison of the step words gives Hamming distance $2n-6$.

It remains to establish the stronger initial-level statement. With one E-block, alternation forces the unique block word ED^{m-1} and score P_{m+1}. A three-E word ending in E has the form

$$A_r = ED^{2r}ED^{m-2r-3}E, \qquad 0 \le r \le n-3.$$

Indeed, the endpoint counts force the movable E-position to be odd, and a direct prefix check shows that every displayed word decodes to a path in $\mathcal{D}(n, n-1)$. Equation (7) gives

$$M(A_r) = P_{m+1} + f(2r+1).$$

The first determinant identity in the proof of Lemma 7.1 shows that

$$f(1) > f(3) > \cdots > f(m-3).$$

Thus, in decreasing order of r, the A_r are the $n-2$ singleton levels immediately above the one-E minimum, and $A_0 = X_n$ is the largest of them.

For a three-E word ending in D, equality in the lower estimate $M \ge M(Y_n)$ can occur only for the position pairs $(2,4)$ and $(m-5, m-3)$. The first decodes to a word beginning RRUU and violates the rational-Dyck inequality at that prefix. The second decodes to Y_n. The strict replacement argument already shows that every word with at least five E-blocks lies above this level. Hence Y_n is the unique next path. This completes the proof of all assertions in Theorem 1.5.

8. Computational verification and formalization status

All computations used in the paper are finite and exact. The two paths in Theorem 1.2 are checked by direct prefix tests, integer matrix multiplication, and a graph-intrinsic dihedral invariant. The bounded claim in Proposition 4.1 is checked by two independent enumerations. The best-first construction and representative prefix certificates have also been compared with exhaustive enumeration on finite ranges. Every comparison uses integers or reduced rational squares.

The finite statement in Proposition 4.1 is confined to total length at most 25. In contrast, Theorems 1.1, 1.3, 1.4, and 1.5 are uniform mathematical statements proved in the preceding sections.

The accompanying Lean 4 development uses Mathlib [3, 6]. It formalizes the rational-Dyck carrier, both exact matrix scores, and the literal cyclic square-band quotient. It proves that every chordless four-cycle of the band is a tile, that the shared-edge graph of the tiles is the required cycle, and that its adjacent-edge decoration is exactly the cyclic equality/difference code. This yields Theorem 1.1 on the literal quotient graph. Burnside counting of the resulting involution, including the odd- and even-reflection fixed-point counts, gives Corollary 3.4 and its Catalan specialization.

For the counterexample, Lean checks the two carrier conditions, the displayed identity for the square of the literal real periodic Lagrange value, and nonisomorphism of the literal quotient bands. It also proves the minimality theorem for every coprime endpoint of total length at most 25. The finite search contributes ordinary data only: the kernel recomputes all 934,635 path assignments, the 471,455 exact score classes, the 22,569 prefix regions, and the complete endpoint range.

For the cover problem, the development defines the actual frontier state and its two transitions: erase and expand a selected nonleaf by precisely its feasible children, or erase and append a selected leaf to the output. It proves preservation of the frontier partition, antichain, score-order, and completeness invariants. The potential $\sum_\alpha 3^{a+b-|\alpha|}$ strictly decreases at every transition, so the traversal terminates with the output a permutation of all carrier paths. This constructs complete best-first runs for both manuscript keys and proves their score order and equality-batch completeness. The development also proves the consecutive-level and complete-bipartite characterizations, both prefix-antichain equivalences, the matching and Lagrange prefix bounds, the endpoint-parity and unit-gap results, all four local-interchange identities, the Pell extremal lemma, and the complete nonlocal family of Theorem 1.5. The large minimality certificate is split into independently compiled class, assignment, compact-frontier, and endpoint modules. The publication root, the exact manuscript formula map, the theorem map, and the axiom report are compiled with `-trust=0` under a 32GB aggregate memory limit.

The development has no project axiom, opaque theorem, native-evaluation bridge, or published-result premise. It defines a square snake as an explicit finite vertex-and-edge type and a perfect matching as an edge selection meeting every vertex exactly once; local equivalences prove the two-state recurrence above. It also defines the periodic Lagrange value by the standard maximum of forward-plus-backward cut values, proves that the backward value is the negative conjugate, and derives equation (1). The axiom report for every publication endpoint contains only Lean/Mathlib's standard `propext`, `Classical.choice`, and `Quot.sound`.

No theorem-valued certificate or search conclusion is assumed. The same is true of every path-code translation, graph isomorphism, cover implication, counting identity, and final composition. Every numbered result has an explicit entry in the accompanying manuscript-to-Lean map. The reproducibility documentation records the exact commands, generated-source counts, formula and theorem maps, and trust report.

Data and code availability. The Lean source, exact certificates, verification scripts, manuscript, and camera-ready PDF are available at `https://github.com/crabsatellite/lattice-path-orders`; versioned releases are archived through Zenodo.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI Codex and an author-developed local research system for proof development, code, copy-editing, and the Lean 4 formalization. The author checked the claims and citations, matched the manuscript statements to the Lean declarations, and reran the finite checks, formal build, and axiom audit. AI output is not mathematical evidence; the author is responsible for the definitions, proofs, citations, code, and manuscript. Further technical details will be disclosed separately.

Acknowledgements

The author thanks the developers of Lean and Mathlib for the formalization environment.

References

1. P. J. Apruzzese and Kevin Cong, *On two orderings of lattice paths*, 2023.
2. İlke Çanakçı and Ralf Schiffler, *Snake graph calculus and cluster algebras from surfaces III: band graphs and snake rings*, International Mathematics Research Notices (2019), no. 4, 1145–1226.
3. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction—CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
4. Gerhard Ramharter, *Extremal values of continuants*, Proceedings of the American Mathematical Society **89** (1983), no. 2, 189–201.
5. Ralf Schiffler, *Perfect matching problems in cluster algebras and number theory*, Open Problems in Algebraic Combinatorics, Proceedings of Symposia in Pure Mathematics, vol. 110, American Mathematical Society, Providence, RI, 2024, pp. 361–371.
6. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, ACM, 2020, pp. 367–381.

Independent Researcher, Tokyo, Japan; ORCID 0009-0008-4516-8946
Email address: `cl7076@nyu.edu`